

SLABS: An improved probabilistic method to assess the avalanche risk on backcountry ski tours

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ABSTRACT

The main danger to which backcountry skiers are exposed are dry slab avalanches triggered by skiers themselves. In this paper we present a new probabilistic method to assess the avalanche risk when planning a ski tour: SLABS (Screening the Likelihood of Avalanches on Backcountry Ski tours). The SLABS method is the first fully statistically derived probabilistic method for backcountry skiing. We used a data set with GPS (Global Positioning System) tracks of backcountry ski tours (57.8 thousand km) and a data set of 1250 accidents recorded in Switzerland during the last 20 winter seasons. A GAM (Generalised Additive Model) with binomial link discriminating between accidents and non-accidents was fitted to these data. As predictors the model uses a non-linear relation with the slope angle, and a linear relation with the danger level, the elevation and the aspect. Conditions with elevated and high risk were defined such that the accident prevention rate is 80% and 60%, respectively. A comparison of the SLABS method with the Graphical Reduction Method (GRM), the Professional Reduction Method (PRM) and the Quantitative Reduction Method (QRM) shows that it offers more freedom of movement for a given accident prevention rate. It is the first time that probabilistic methods are compared in terms of their trade-off between accident prevention and freedom of movement. Because the SLABS method is not suited for mental arithmetic it is implemented in the website www.skitourenguru.ch. This website evaluates on a daily basis the avalanche risk along thousands of backcountry tours throughout the Alps to help skiers plan their next tour. For a single slope the risk assessment of SLABS provides a starting point that has to be supplemented with local observations.

1. Introduction

When travelling in the backcountry in winter the main danger for skiers are dry slab avalanches triggered by the skiers themselves. To assess the risk many decision-making frameworks (DMFs) are available. Landrø et al. (2020b) give an overview of the most commonly used DMFs and distinguish two main categories: probabilistic and analytic methods. Probabilistic methods combine information about the current conditions and the terrain into a risk score. The main inputs are the avalanche forecast (e.g. danger level) and terrain properties (e.g. slope angle, aspect, elevation). When the risk score exceeds a predetermined threshold, extra caution or return is recommended. Probabilistic methods are easy to use but give a rather general assessment that might not always be valid for a specific slope. On the other hand, analytical

methods assess the avalanche risk mainly with local observations of danger signs (e.g. recent avalanches, shooting cracks or whumpf sounds), snow profiles and stability tests (e.g. Extended Column Test (Simenhois and Birkeland, 2006) and the Rutschblock test (Föhn, 1987)). Analytical methods give potentially a more locally valid assessment but danger signs are not always present or noticed and stability tests are time consuming. In practice a mix of probabilistic and analytic methods is recommended, even if some people tend to rely more on the probabilistic methods while others rely more on the analytical methods. Landrø et al. (2020a) did a survey among mountain professionals and concluded that they rely more on analytical methods. Often it is recommended that beginners stick to the recommendations of probabilistic methods supplemented as far as possible with local observations.

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Also in avalanche research machine learning techniques are catching on. Just a few examples: Pérez-Guillén et al. (2022) trained a random forest model with meteorological and snow cover simulations to predict the avalanche danger level; Mayer et al. (2022) used the same technique to train a model with simulated snow stratigraphy to predict snow instability; and Dkenge Sielenou et al. (2021) used a random forest to predict the avalanche activity from the weather forecast to issue warnings for natural avalanches endangering people, properties and infrastructure. In this paper we bring these novel techniques to the field of probabilistic methods. Up to now, these methods are mainly based on accident analysis, avalanche knowledge and experience. They are not or only partially underpinned by a statistical data analysis. The objectives of this paper are 1) to investigate in a systematic way how the avalanche forecast and terrain properties influence the probability to trigger a dry slab avalanche (For the purposes of this paper, a dry slab avalanches will be referred to simply as 'avalanche'.) 2) to develop a new, data based probabilistic method and 3) to compare the performance of the new method with the existing ones. To this end we use a large set of backcountry travel data and accident data; the Avalanche Risk Property Data set or ARPD (Schmudlach, 2021a; Winkler et al., 2021). These data are used to determine the parameters of the significant *covariates* or *predictive variables* of a statistical model. This statistical model is the basis of a new probabilistic method that we called SLABS (Screening the Likelihood of Avalanches on Backcountry Ski tours). Schmudlach et al. (2018b) already used the ARPD to develop a probabilistic method: the Quantitative Reduction Method (QRM), explained in Appendix A.1. The difference is that the present analysis uses the raw input from the avalanche forecast and terrain properties instead of two indicators that combine several raw inputs. Winkler et al. (2021) used the ARPD to derive risk ratios for the danger level, critical elevations and aspects. In the present study we combine all relevant covariates in one statistical model. In this way risk ratios for each covariate can be calculated taking into account that other covariates change at the same time. To compare the new method with the existing methods, all methods are applied to the ARPD. We assess their performance in terms of the trade-off they offer between accident prevention and freedom of movement. This is only possible because the ARPD contains accidents and travel data (i.e. non-accidents). To our knowledge it is the first time that such a comparison is performed. Probabilistic methods have been compared by other authors (McCammon and Haegeli, 2007) but only in terms of prevention rates. Such an analysis ignores the fact that probabilistic methods are not perfect, they often advice against a route even when no avalanche would be triggered. Imagine a probabilistic method that recommends to go on a ski tour only when the danger level is 1-Low. This method would have an accident prevention rate of about 95%. But with the knowledge that only about 25% of ski tours occurs at danger level 1-Low this method drastically restricts the freedom of movement.

Both the travel and accident data are only available for Switzerland. Therefore we will limit the comparison to the most common probabilistic methods used in Switzerland: the Graphical Reduction Method (GRM), the Professional Reduction Method (PRM) and the Quantitative Reduction Method (QRM). The GRM and PRM are established methods for more than 20 years. Both are hands-on tools that backcountry recreationists can apply on their own and can also be used to teach avalanche risk assessment principles. The QRM is more recent. It is used by the website www.skitouren.guru.ch since winter season 2018–2019 to provide a daily risk assessment for backcountry ski tours based on the most recent avalanche forecast. Readers who are not familiar with these probabilistic methods are referred to Appendix A.1.

2. Description of the dataset

The model proposed in this paper is based on the Avalanche Risk Property Data set (ARPD). The ARPD consists of a set of covariates attached to travel points and accident points. The accident points serve

as event-points, the travel points as non-event-points. A technical description of the ARPD is provided in Schmudlach (2021a). A summary of the the ARPD can be found in Winkler et al. (2021).

2.1. Backcountry travel data

Travel data about backcountry skiing was provided by the online platforms www.gipfelbuch.ch, www.camptocamp.org and www.skitour.guru.ch. The original GPS data were processed according to the following procedure:

1. GPS tracks that followed for more than 70% of the total length a road or driveable track were discarded.
2. A specially designed algorithm detected and removed GPS artefacts (like GPS spikes).
3. The GPS tracks were re-sampled with 10 m distance.
4. The tracks were converted to a point collection.
5. All points outside Switzerland were discarded.
6. All travel points closer than 200 m to a ski slope were removed to avoid off-piste skiing close to the slopes being included.
7. If no avalanche forecast from the previous evening could be found for a particular point, it was discarded.
8. It was verified that each GPS point in terms of coordinates and time stamp is only contained once in the data set.

Travel data are available for the winter seasons from 2005–2006 to 2020–2021. The total data set contains 5.78 million travel points, 57.8 thousand kilometres and 8558 tours. We estimate that this represents about 0.5 per thousand of the backcountry travel in the period from season 2001–2002 to season 2020–2021 (the period for which also accidents are available). The number of tours for each year is estimated by linear interpolation between the years 1999, 2007, 2013 and 2020. For the first three years the estimate is taken from a paper by Winkler et al. (2016) who estimated the amount of ski tours in the years 1999, 2007 and 2013 (630, 780, and 850 thousand tours, respectively). The amount of tours in 2020 was estimated applying the change between 2014 and 2020 from surveys on sports practice in Switzerland (Lamprecht et al., 2015; Bürgi et al., 2021).

Crowd sourced data is potentially subject to a sample bias. In Winkler et al. (2021) and Schmudlach et al. (2018b) this issue is discussed in depth. However, a model based on accidents alone shows bigger errors in risk estimates than one based on accidents and eventually biased travel data.

2.2. Accidents

In Switzerland, the WSL Institute for Snow and Avalanche Research (SLF) maintains a database including all reported avalanche accidents. This data has been used in numerous studies to explore accident patterns. We considered only accidents that comply with the following criteria:

- Winter seasons from 2001–2002 to 2020–2021.
- At least one person was caught.
- The accident record is marked in the database as reliable and accurate.
- An avalanche forecast from the previous evening is available.
- The reported activity is backcountry skiing or snowboard touring (in contrast to off-piste skiing with mechanical ascending devices).

An avalanche record usually only contains the avalanche release point and not the trigger point. The release point is the highest point of the release area, while the trigger point is the position of the skier at the moment the avalanche was triggered. Most avalanches are triggered from the release area. Harvey et al. (2018) computed an average release area size of 2520 m² for 5225 small and medium-sized avalanches

recorded in the region around Davos. Depending on the release area shape this area corresponds to a release area length in the range of 40 to 80 m. In order to create possible trigger points we place 7 equidistant points on a 60 m long downhill trajectory starting from the release point. In this way most of the possible trigger points are placed within or just below the release area. In [Winkler et al. \(2021\)](#) the bias of the accident records towards accidents with heavy consequences is discussed. The latter are more likely to be reported.

2.3. Covariates

For each point in the ARPD a set of covariates was collected. This section describes the covariates that are eligible for the model and some others that are used for model validation. The covariates refer either to the terrain, to the avalanche forecast or to both. All terrain related covariates were calculated from the Digital Elevation Model *swissAL-TI3D* ([Swisstopo, 2018](#)) with a resolution of 10 m and approximate accuracy of 1 m.

- **Slope Angle (SA):** The local slope angle is calculated with the GDAL tool *gdaldem* ([GDAL/OGR contributors, 2021](#)). Using the local slope angle might introduce an undesired bias. When a skier walks up a slope the least steep sections in the slope are preferred. The 7 points placed on a downhill trajectory are potentially steeper than travel points of backcountry skiers. A model using the local slope angle will exploit this bias to distinguish accidents from travel points. To avoid this problem other slope angle covariates are introduced that take into account a larger area around each point.
- **Maximum Slope Angle (MSAx) with $x \in [40, 70, 100, 150]$:** MSAx is the 85th percentile of the local slope angles on the Relevant Slope Area (RSA) of a specific location. The parameter x is a measure for the size of the RSA in metre. In [Appendix A.3](#) it is explained how the RSA for a particular location is delineated. The aim is to consider the slope properties in a wider environment around the skier's position. If the values of x are too small, there is the same danger of bias as with the slope. We therefore calculate the MSAx for four characteristic radii x equal to 40, 70, 100 or 150 m.
- **Danger level dependent Maximum Slope Angle (SA_{jx} and SA_p):** Because it is common practice to consider a larger slope area at higher danger levels (e.g. GRM and PRM), these covariates combine SA at low danger levels and MSAx at higher danger levels. This was implemented with a jump (SA_{jx}) and with progressively (SA_p). The covariates SA_{jx} use SA up to 2-Moderate and MSAx from 3-Considerable. The covariate SA_p uses SA at 1-Low, MSA40 at 2-Moderate, MSA70 at 3-Considerable and MSA100 at 4-High.
- **Terrain Indicator (TI):** TI is a value between 0 and 1, derived from the slope features on the Relevant Slope Area (RSA). In addition to the slope angles on the RSA, TI considers also the size of the RSA. It indicates how appropriate the terrain is to trigger an avalanche. For more details see [Schmudlach et al. \(2018b\)](#). Points with a terrain indicator greater than 0.25 are considered *avalanche terrain*. 42% of the travel points and 95.3% of the accidents are located inside *avalanche terrain*. The TI is used by the QRM and in [Winkler et al. \(2021\)](#).
- **Aspect (ASPECT):** Slope aspect calculated with the GDAL tool *gdaldem* ([GDAL/OGR contributors, 2021](#)). Slopes with aspect 0° and 90° face North and East, respectively. The cosine of the ASPECT was included as the covariate NORTH and the sine as the covariate EAST. They express how Northerly or Easterly an aspect is.
- **Elevation (ELE):** The elevation above sea level of the point in meters.
- **Plan Curvature (PLANC):** The plan curvature, calculated with the GRASS tool *r.param.scale* ([GRASS Development Team, 2017, 2021](#)).
- **Terrain Fold (FOLD):** FOLD visualises ridges, valley bottom lines, hillside toes or slope edges. FOLD is defined by the maximum angle between 5 pairs of normal vectors placed oppositely on a circle with

radius 10 m around the point. More details you find in [Schmudlach et al. \(2018b\)](#).

- **Distance to Ridge (DIST_RIDGE):** The distance in meters to the nearest ridge.
- **Maximum Fall-down Velocity (FD_MAXV):** The maximum velocity skiers would reach in case they fall down from the respective point. The value is derived from a simplified physical model. More details you find in [Harvey et al. \(2018\)](#). FD_MAXV is correlated to all slope angle covariates.
- **Forest Density (FD):** Tree cover density according to Corine Land Cover data ([ESA, 2017](#)). FD is expressed in percent.

A second set of covariates is derived from the avalanche forecast issued by the Swiss avalanche warning service (SLF) at 17:00 h the day before:

- **Danger Level (DL):** The danger level provided by the avalanche forecast: 1-Low, 2-Moderate, 3-Considerable, 4-High. Level 5-Very High is too rare and thus not considered.
- **Critical Aspects (CA):** Critical aspects as indicated in the avalanche forecast, 8 sectors of 45°.
- **Critical Elevation (CE):** Critical elevations as indicated in the avalanche forecast.
- **Avalanche Problems (AP):** Avalanche problems like fresh snow, wind-drifted snow, old snow, glide snow or wet snow. This information is only available from winter season 2012–2013 onwards.

The danger level is valid for locations inside the critical elevations and aspects. These locations are also called the core zone. For DL 1-Low, no core zone is specified in the Swiss avalanche forecast, which according to the interpretation aid ([WSL Institute for Snow and Avalanche Research SLF, 2021](#)) means that all aspects and elevations are equally affected. [Winkler et al. \(2021\)](#) have shown that this is not the case and have determined the following risk-based values for critical elevation and critical aspects for DL 1-Low: critical aspects from West over North to South-East and critical elevations above 2000 m. In this paper we use the same definition at level 1-Low.

A third and last set of covariates is derived from combining the avalanche forecast with terrain features:

- **Delta Critical Elevation (DCE):** The elevation difference between ELE and CE. DCE tells how deep the point is inside the core zone in terms of elevation. DCE is positive inside the critical elevations and negative outside.
- **Delta Critical Aspect (DCA):** The minimum angular difference between the aspect of the point and the borders of the critical aspects from the avalanche forecast. DCA tells how deep the point is inside the core zone in terms of aspect. Aspects inside the critical aspects are positive.
- **Inside Critical Elevations (INCE):** A binary variable that indicates whether the point is located inside (1) or outside (0) the critical elevations.
- **Inside Critical Aspects (INCA):** A binary variable that indicates whether the point is located inside (1) or outside (0) the critical aspects.
- **Aspect Overlapping Fraction (AOF):** The fraction of the range of aspects in the relevant slope area (RSA) that overlaps with the critical aspects as indicated by the avalanche forecast. Only aspects with slope angles $> 25^\circ$ are taken into account. 0 means that all aspects in the RSA are outside the critical aspects, 1 means the all aspects in the RSA are inside. AOF is a blurred version of INCA. A detailed description is available in [Appendix A.3](#)

In [Appendix A.2](#) some additional information about the covariates is provided: summary statistics, probability density plots and histograms. The next section describes how this data set is used to develop a new

probabilistic method: SLABS.

3. Methodology

This methodology section consists of three parts. The first and most extensive one describes the *statistical model* that we fit to the accident and travel data. We have to make sure that the model is not fitted to some peculiarities of this data set. This can be achieved by splitting the data set in two parts: one to train the model and another to validate it. Only covariates that significantly improve predictions in both the training and the validation data sets are used in the final model. The second part of this section describes how the output of the statistical model, an accident probability, can be transformed in a *probabilistic method* giving backcountry travel advice. There are no objective criteria to do this. The amount of risk one is willing to take is in the end a personal choice. But by setting a goal on the share of accidents we would like to avoid, limits can be set. The last part of this section discusses how to *compare different probabilistic methods*. Each method is applied to all travel and accident points of the data set and gives a risk assessment. In a next step we check how many accidents are correctly detected and how many travel points are by mistake labelled as accident. A good method identifies many accidents without making too many mistakes. The rest of this section explains the methodology in more detail.

3.1. The statistical model

In the development of the model the following steps were followed:

1. Choice of an appropriate statistical model
2. Preparation of data sub-sets with independent observations.
3. Step-wise model selection with K-fold cross-validation.
4. Pearson residual analysis on the final model.

The objective of the model is to predict an accident probability as a function of several continuous and categorical covariates or predictive variables. In this case we cannot use a multi-variate linear regression. The result, a probability, can only be a number between 0 and 1 while a linear combination of parameters could yield values outside this interval. Therefore the probability is transformed. We model the natural logarithm of the odds-ratio or logit of the probability. The logit of a probability p is defined as: $\text{logit}(p) = \ln\left(\frac{p}{1-p}\right)$. The values of $\text{logit}(p)$ lie between $-\infty$ and ∞ . The relation between some covariates and $\text{logit}(p)$ can be much more complex than a straight line. E.g. it is known that in flat terrain avalanche triggering is rare while it is much more frequent in slopes steeper than 30° . To model such non-linear dependencies splines, called smoothers in this context, are appropriate. A model that combines a logit link function with smoothers of the covariates is called a Generalised Additive Model (GAM). To fit the GAM the *mgcv* package available in R (R Core Team, 2021; Wood, 2017) was used.

When fitting a statistical model on observations, a fundamental assumption is that these observations are independent. This is clearly not the case for the ARPD. The sampling of the travel data at 10 m intervals and the 7 points on a downhill trajectory for each accident cause dependency in the data. Nearby points are very similar and are almost copies of each other. This effect is called pseudo-replication. To create a sub-set of independent travel data we proceeded with an iterative process. The final GAM is applied to all the travel points and the auto-correlation is calculated for different spatial lags. The spatial lag that results in an average auto-correlation below 20% is 200 m. As a rule of thumb an auto-correlation below 20% is considered appropriate by Zuur (2017). Thus we will only use points every 200 m or 5% of the original travel data, sampled at 10 m. The chosen 200 m exceed the median width (50 m) and length (80 m) of the release area of a human-triggered avalanche (Schweizer and Lüscher, 2001). To create independent accident data a different strategy was used. The accidents are independent

but the 7 points on a downhill trajectory are not. Therefore 7 accident data sets were created by sampling randomly one of the 7 points of each accident. No repetitions were allowed.

We are interested in a model that can be used for predictions and not so much in a descriptive model of our data. Therefore the data will be split in two parts: one for training and another for validation. Before splitting up the data, the minimum amount of travel and accident points was determined with a sensitivity study. Around 50,000 travel points or 0.87% of the travel data are needed to guarantee stable parameter estimates. In terms of accidents at least 65% of the data (about 800 of the 1250 accidents) are needed. These sensitivity studies are carried out with the final model in an iterative process.

After the preparatory steps described above the cross-validation scheme can be defined. The limiting factor is the amount of accidents needed to fit a model. We chose to carry out a 5-fold cross-validation (Hastie et al., 2009). This means that the accidents are randomly divided in five groups or folds of 250 accidents. Four of the five folds (1000 accidents) are used for training and the other fifth for validation. Also the travel data are divided in five folds of 57800 points. In a typical 5-fold cross-validation four folds would be used for training and one for validation. In this case it is enough to take only one fold of travel data for training. This makes the training and validation data more independent than usual. This procedure is repeated five times to obtain five folds. The particularity of this data set is that for each accident there are 7 points. This allows to split up each of the five folds into seven sub-folds. These sub-folds are constructed by randomly selecting one point from each accident. Each model was trained and validated on these 35 sub-folds. Fig. 1 illustrates the construction of the data sets for 5-fold cross-validation.

With each fold a forward step-wise model selection process is carried out. We start with all possible univariate models on each training data set. The model is applied to the corresponding validation data set. For each model the average Bayesian Information Criterion (BIC) over the five folds corresponding to the 7 accident sub-sets is calculated with Eq. (1).

$$\bar{BIC} = \frac{1}{M} \frac{1}{K} \sum_{m=1}^M \times \sum_{k=1}^K \left[-2 \log \left(\prod_{i=1}^{N_{k,m}} p_i^{a_i} (1-p_i)^{(1-a_i)} (1-a_i) \right) + d \log(N_{k,m}) \right] \quad (1)$$

Where $K = 5$ and $M = 7$ are the number accident sub-sets and folds, $N_{k,m}$ is the number of observations in fold k with accident sub-set m , p_i is the avalanche probability for observation i , a_i is the outcome of observation i (1 for an accident, 0 for a travel point) and d is the effective number of degrees of freedom of the model.

First all univariate models are fitted. The model with the lowest $\bar{BIC}$ is the starting point for the next step of the model selection. Covariates

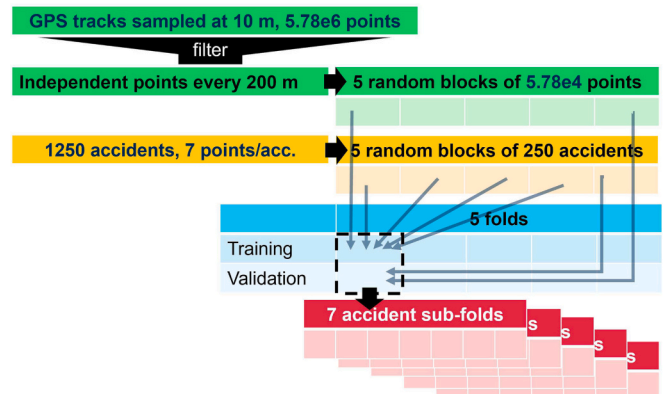


Fig. 1. Overview of the construction of data sets for 5-fold cross-validation.

are added until there is no improvement in the BIC anymore. However, there are some other considerations. When a covariate with smoother is only slightly better than the linear covariate it is better to include the latter in order to limit the degrees of freedom of the model.

To produce the final model the 7 points of all accidents and all travel data sampled at 10 m were used. From this model only the model coefficients and the smoothers are used. The standard errors on coefficients and smoothers are under-estimated. Therefore the standard errors are calculated as the average over 7 models using only one of the 7 accident points per accident and travel data sampled at 200 m.

In a final validation the normalised Pearson residual of covariates that are inside and outside the model and their interactions are calculated with Eq. (2) and checked for the absence of trends (Zuur, 2012).

$$r_I^p = \frac{y_I - \mu_I}{\sqrt{\mu_I}} \text{ with } \mu_I = \sum_{i \in I} p_i \quad (2)$$

First the range of a covariate or interaction is divided in a number of intervals. For each interval, I , y_I is the observed number of accidents and μ_I is the expected number of accidents. It is calculated with the model as the sum of the accident probability of all observations in the interval, I . Since count data follow a Poisson distribution the expected value equals the variance. The difference between the number of observed and expected accidents divided by the square root of the expected number of accidents should be normally distributed, should lie mainly between -1 and 1 , and should show no trends with the binned covariate. In this way the model residuals can be evaluated in the same way as for a generalised linear model with Poisson link function (Zuur, 2012).

The model gives the expected value and the standard error of the logit of the accident probability ($\text{logit}(p)$) for a combination of covariates. The absolute value of this probability is not meaningful because of under-reporting: we have a high percentage of accidents and only a small fraction of the travel data. Hence, the average accident probability of the model is too high. This is not a problem because we will only use the model to calculate *relative risks* or *risk ratios* (RR) between different conditions (combinations of covariates). The relative risk is often used in a medical context and compares the risk of exposed and unexposed subjects (Yada, 2017). Here we compare different conditions: e.g. the relative risk between two danger levels or two different slope angles. The risk ratio is calculated dividing the accident probabilities under both conditions. The probabilities of the compared conditions are obtained by taking the inverse logit (Eq. (3) with $x = \text{logit}(p)$) of the model output (see Eq. (4) further below).

$$\text{logit}^{-1}(x) = \frac{\exp(x)}{1 + \exp(x)} \quad (3)$$

3.2. From the statistical model to the probabilistic method

The output of the statistical model is the probability that a point in the data set is an accident as a function of the selected covariates. To derive travel advice from it, limits for these probabilities have to be defined. There are two ways to do this: 1) Based on desired accident prevention rates. The advantage of this approach is that it offers a clear message: by avoiding particular points, a particular share of accidents can be prevented. The drawback is that these limits change when the data set changes. 2) Another way to define limits is to consider a specific combination of conditions as too dangerous. E.g. Munter (Munter, 2017) defined the combination of danger level 3-Considerable, a slope angle of 40° and North aspect as the *triangle of death*. With the model we could calculate the corresponding probability and avoid all conditions that have the same or a higher accident probability. Other limits could be the ones proposed by the GRM at level 3-Considerable: the risk is considered *high* on slopes steeper than 35° and *elevated* on slopes steeper than 30° .

For the SLABS method we chose the first approach. We set the limit for *high* risk such that 60% of the accidents can be prevented and the limit for *elevated* such that 80% of the accidents can be prevented. The

same limits were used in the QRM. For lower accident probabilities the risk is considered *slight*. The numbers correspond to typical prevention rates measured by McCammon and Haegeli (2007). There are no objective criteria to set such limits. The amount of risk one is willing to take is in the end a personal choice.

3.3. Comparing probabilistic methods

After having developed the SLABS method we want to compare it to the GRM, PRM and QRM. For this each method is applied to all travel and accident points of the data set to give an indication of the risk. (e.g. the categories slight, elevated or high for the GRM or the residual risk value, calculated with formula A.1, for the PRM). Then we check for each category or value how many accidents are correctly detected and how many travel points are by mistake labelled as an accident. The corresponding fractions of accidents and travel points are calculated by dividing them by the total number of accidents and travel points, respectively. These two fractions are called True Positive Rate (TPR) and False Positive Rate (FPR). Where a *positive* result means an accident. The TPR is the accident prevention rate. The FPR is the share of the terrain that has to be avoided to achieve a desired prevention rate. It is a measure for the restriction of the freedom of movement. A plot of the TPR against the FPR is called the *Receiver Operating Characteristic* (ROC) (Fawcett, 2006). We would like to have a ROC that has a high TPR for small FPR, or high prevention rates for few restrictions of movement. This means that the area under the ROC should be big. The *Area Under Curve* (AUC) is a commonly used quality indicator of binary classification models. A model with an AUC of 1 distinguishes perfectly between accidents and non-accidents while a model with an AUC of 0.5 is no better than random. This will be the criterion to compare the methods. For each accident all 7 points are used because the dependency between points has no influence on the ROC.

4. Results

4.1. A Generalised Additive Model to assess avalanche risk ratios

The step-wise selection resulted in the following model for the logit of the probability on an accident (p):

$$\text{logit}(p) = \beta_0 + s_1(\text{MSA40}) + \beta_1 DL + s_2(\text{DCE}) + \beta_2 \text{AOF} \quad (4)$$

β_0 , β_1 and β_2 are the intercept, the coefficient of DL and the coefficient of AOF , respectively. $s_1()$ and $s_2()$ are the smoothers of MSA40 and DCE , respectively. The relation between the $\text{logit}(p)$ and each covariate is shown in Fig. 2. The distance between the horizontal dashed lines corresponds to the logit of the risk ratio between two consecutive danger levels. Table 1 lists some selected risk ratios. The covariates are now discussed in the order they were selected:

1. A smoother $s_1()$ for MSA40 . The default smoother type of the R-mgcv package was used: a thin plate spline. The effect of the type of smoother was assessed but had no significant impact on the result (see Appendix A.5). The smoother has 6.79 effective degrees of freedom. The local slope angle (SA) was selected first in three of five folds (averaged over the 7 accident sub-sets), MSA40 was first in the two other folds. However, because SA exploits an artefact (as explained before) it was disallowed and the second best covariate, MSA40 , was chosen instead. The other maximum slope angles (MSAx with $x \in [70, 100, 150]$) scored worse. Trials with (SA_jx) and (SA_p) showed that they incorporate part of the effect of danger level. As a consequence, the difference between danger levels 2-Moderate and 3-Considerable becomes insignificant. In other words, there is a high and not desirable correlation between these slope angle covariates and the danger level. (SA_jx) and (SA_p) were therefore not used. TI is similar to MSAx , however priority was given to MSAx as its more intelligible. Fig. 2a shows the smoother of MSA40 . The logit of the

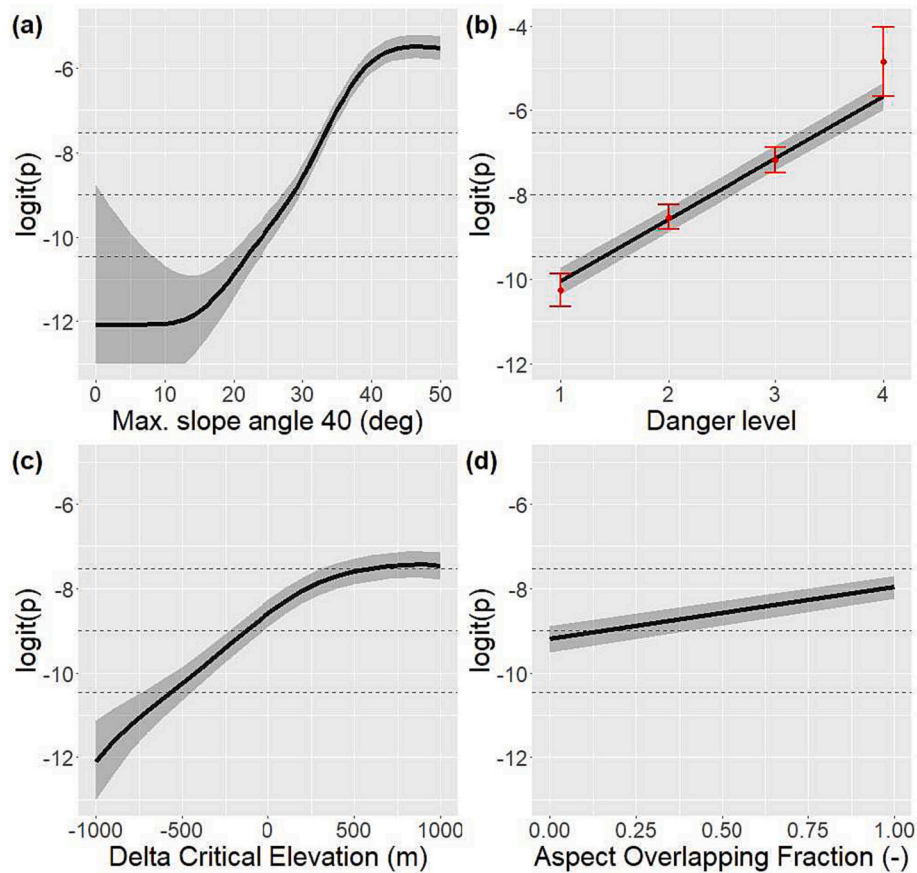


Fig. 2. $\text{logit}(p)$ as a function of single covariates with 95% prediction confidence intervals. All plots have the same Y-axis range. The horizontal dashed lines correspond to the difference between two consecutive danger levels. a) Smoother of MSA_{40} . b) Continuous and categorical (red) danger level DL . c) Delta Critical Elevation, DCE . d) Aspect Overlapping Fraction (AOF). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1

Selected Risk Ratios and their 95% confidence interval calculated with model of Eq. (4), only the risk ratios for DL cat. were calculated with a model with DL as a categorical covariate.

label	value	ref.value	RR	CI
MSA_{40}	25	20	2.9	[1.9, 4.4]
MSA_{40}	30	25	3.3	[2.5, 4.3]
MSA_{40}	35	30	4.9	[4.1, 5.8]
MSA_{40}	40	35	3.2	[2.8, 3.6]
MSA_{40}	45	40	1.4	[1.3, 1.6]
MSA_{40}	50	45	1.0	[0.9, 1.1]
DL cont.	2	1	4.3	[3.9, 4.7]
DL cont.	3	2	4.3	[3.9, 4.7]
DL cont.	4	3	4.3	[3.9, 4.7]
DL cat.	2	1	5.7	[4.3, 7.3]
DL cat.	3	2	3.9	[3.4, 4.4]
DL cat.	4	3	10.1	[4.3, 20.2]
DCE	0	-500	5.3	[3.9, 7]
DCE	500	0	2.7	[2.3, 3.1]
DCE	1000	500	1.1	[0.9, 1.4]
AOF	1	0	3.4	[2.7, 4.2]

accident probability, $\text{logit}(p)$, increases strongly with slope angle, especially between 30° and 40°. The risk ratios between 35° and 30° and between 40° and 35° are 4.9 [4.1, 5.8] and 3.2 [2.8, 3.6], respectively (Table 1). Above 40° the increase of $\text{logit}(p)$ flattens off. Above 45° there is a slight but insignificant descent. The risk ratio between 50° and 45° is 1.0 [0.9, 1.1]. Also the $\text{logit}(p)$ increase in between 20° and 30° is relatively strong.

2. *The Danger Level, DL .* Although DL is a categorical covariate, it was also included as continuous covariate in the model selection. The continuous covariate resulted in a model with a slightly lower $\bar{BIC}$ and scored best in four of the five folds. DL as categorical covariate had the second lowest $\bar{BIC}$. DL as continuous covariate was included in the model because it has two advantages over the categorical covariate: 1) The estimates for level 4-High are more stable when in the future more data will be available. 2) It is easy to implement the intermediate danger levels (Techel et al., 2020, 2022) that are published by the Swiss avalanche warning service from winter 2022–2023. The risk ratio between two consecutive danger levels is 4.3 [3.9, 4.7] (Table 1). In the model with categorical DL the risk ratio between 2-Moderate and 1-Low is 5.7 [4.3, 7.3], between 3-Considerable and 2-Moderate it is 3.9 [3.4, 4.4] and between 4-High and 3-Considerable it is 10.1 [4.3, 20.2]. The uncertainty on the last one is big because few data are available. Fig. 2b shows the $\text{logit}(p)$ as a function of DL for the categorical and continuous case. The differences between both models are small.
3. *A smoother $s_2()$ for DCE .* Also here a thin plate regression spline was used as smoother. The impact of the type of spline was assessed but negligible (see Appendix A.5). The smoother has 6.78 effective degrees of freedom. This smoother rises almost linearly up to 500 m above the critical elevation, then it flattens as shown in Fig. 2c. Table 1 shows that at the CE the risk is 5.3 [3.9, 7] times higher than 500 m below the CE. This increase continues above the CE; 500 m above the CE the risk is 2.7 [2.3, 3.1] times higher than at the CE. Then the rise flattens off. The rise from 500 to 1000 m above is not significant (1.1 [0.9, 1.4]).

4. *Aspect Overlapping Fraction, AOF*. In four of the five folds the covariate *AOF* scored best, followed by *DCA* with smoother and *INCA*. *AOF* was included in the model. Compared to the risk ratio between danger levels its effect is smaller (Fig. 2d). Inside critical aspect the risk is 3.4 [2.7, 4.2] times higher than outside (Table 1). The interaction between *DCE* and *AOF* was not significant.

The following covariates were tested but did not result in a model with lower *BIC*: *FOLD*, *PLANC*, *NORTH*, *EAST* and *FD*. *FD_MAXV* and *DIST_RIDGE* were tested but discarded because of their high correlation with *MSA40* and *DCE*, respectively. Also interactions were tested. The interaction between the smoother for *MSA40* and *DL* was not significant, meaning that the same smoother is valid for all danger levels. Also the interaction between *AOF* and *DCE* is not significant. Pearson residuals for the model covariates were checked for the absence of trends and

normality (see Appendix A.7.)

4.2. A new probabilistic method, SLABS

The final model presented in the previous section is combined with accident prevention rates of 60% and 80% for *high* and *elevated* risk, respectively. This provides the basis for the new probabilistic method: SLABS (Screening the Likelihood of Avalanches on Backcountry Ski tours). In the SLABS method two small modifications are applied to the statistical model presented in the section above. The smoothers for *MSA40* and *DCE* decline slightly after 46° and 960 m, respectively. These declines were ignored, the (relative) risk was kept at the maximum value for *MSA40* and *DCE* values above 46° and 960 m, respectively. These declines are not significant and are due to lack of data. We do not see physical reasons for these small declines. Fig. 3 shows the risk

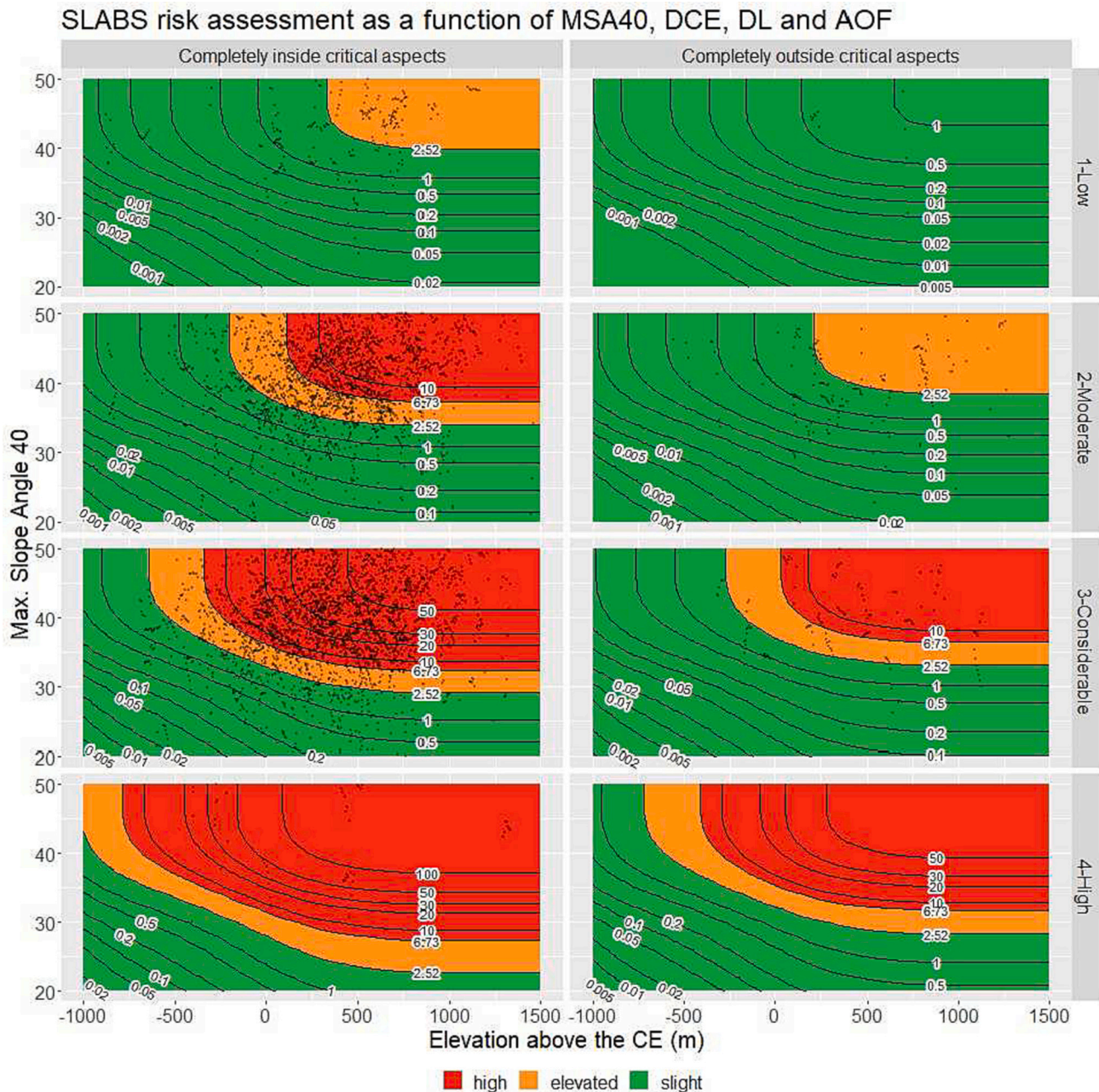


Fig. 3. Risk assessment of the SLABS method as a function of *MSA40*, *DCE*, *DL* and *AOF*. The black lines represent the risk ratio with respect to the average risk. The black dots are the accident points.

assessment by the SLABS method as a function of *DCE* and *MSA40*: red stands for *high*, orange for *elevated* and green for *slight* risk. Each sub-plot applies to a combination of the danger level and AOF (1 or 0, completely inside or outside the critical aspects). The black lines show the risk ratio with respect to the average risk. Also note the large range in relative risks. Combining four covariates it is possible to distinguish between risks that are between 1000 times lower and almost 100 times higher than the average risk. The black dots are the accident points. It is clear from the plots that choosing less steep slopes and staying at lower altitudes decreases the risk ratio. The risk ratio increases strongly at higher danger levels and to lesser extent inside the critical aspects. It is also important to note that there are accident points in the green areas. No method can guarantee absolute safety.

4.3. Comparison between probabilistic methods

Now we compare the GRM, PRM, QRM and SLABS method. As explained in Section 3.3 the Area Under Curve (AUC) of their Receiver Operating Characteristic (ROC) is used. The bigger the AUC, the better a method is in identifying accidents. To calculate the risk scores for the GRM and PRM a choice about the slope angle had to be made. To approximate their use in practice the local slope angle (*SA*), *MSA40*, *MSA70* and *MSA100* were used at danger levels 1-Low, 2-Moderate, 3-Considerable and 4-High, respectively. Schmudlach et al. (2018a) found that *MSA70* corresponds best to what experts consider the relevant slope at level 3-Considerable. In this way the slope angle in a wider area is considered at higher danger levels. In case of the PRM the residual risk for a small group with safety distances ($RF_3 = 3$) was calculated. Fig. 4a shows the ROC for the GRM, the PRM, the QRM and the SLABS method for all points, and Fig. 4b shows the ROC for points in avalanche terrain (i.e. $TI > 0.25$). Table 2 gives the AUC for all points and the ones in avalanche terrain. The former AUCs are higher because non-avalanche terrain contains many additional travel points, but practically no additional avalanche accidents. Travel points are located predominantly at the top right of the ROC. Removing them flattens the ROC and shifts the lower left branch the right. The GRM and PRM have the lowest AUC, the

Table 2

AUC and (FPR, TPR) values at borders between risk levels for each probabilistic method. The Area Under Curve (AUC) should be as big as possible. The False Positive Rate (FPR) corresponds to the *share of activity avoided* and should be as small as possible. The True Positive Rate (TPR) corresponds to the *share of accidents avoided* and should be as big as possible.

Everywhere	GRM	PRM	QRM	SLABS
AUC	0.873	0.858	0.926	0.943
Slight-Elevated	(0.13, 0.84)	(0.31, 0.84)	(0.11, 0.8)	(0.09, 0.8)
Elevated-High	(0.05, 0.6)	(0.04, 0.42)	(0.04, 0.6)	(0.04, 0.6)
Avalanche terrain	GRM	PRM	QRM	SLABS
AUC	0.834	0.812	0.87	0.895
Slight-Elevated	(0.29, 0.87)	(0.41, 0.85)	(0.26, 0.84)	(0.2, 0.83)
Elevated-High	(0.11, 0.63)	(0.09, 0.44)	(0.1, 0.63)	(0.09, 0.62)

QRM scores much better and the SLABS method has the highest AUC. This means that for a given TPR or accident prevention rate the SLABS method restricts the freedom of movement less than the QRM, and much less than the PRM or the GRM.

Table 2 gives an overview of the TPR and FPR at the thresholds used by the different models. The GRM assigns a *slight* risk to combinations of slope angle and danger level in such a way that an accident prevention rate or TPR of 84% is reached but 13% of the travel points have to be avoided. In avalanche terrain the prevention rate is 87% but 29% of the terrain has to be avoided. If only points with *high* risk are avoided the prevention rate in avalanche terrain sinks to 63% but only 11% of the avalanche terrain has to be avoided. The PRM does not make a difference between *slight* and *elevated* risk. We introduced an additional threshold at a residual risk of 1/3. This corresponds to the reduction factor a small group has to reduce the risk by using safety distances. The prevention rate for points with a *slight* risk are similar to the GRM but the FPRs are bigger, so the PRM is more restrictive than the GRM. When only the points with *high* risk are avoided the PRM becomes less restrictive (lower FPR) than the GRM but the prevention rate decreases about 20% points compared to the GRM. This more aggressive strategy is due to the $RF_3 = 3$ for small groups keeping safety distances. The thresholds of the

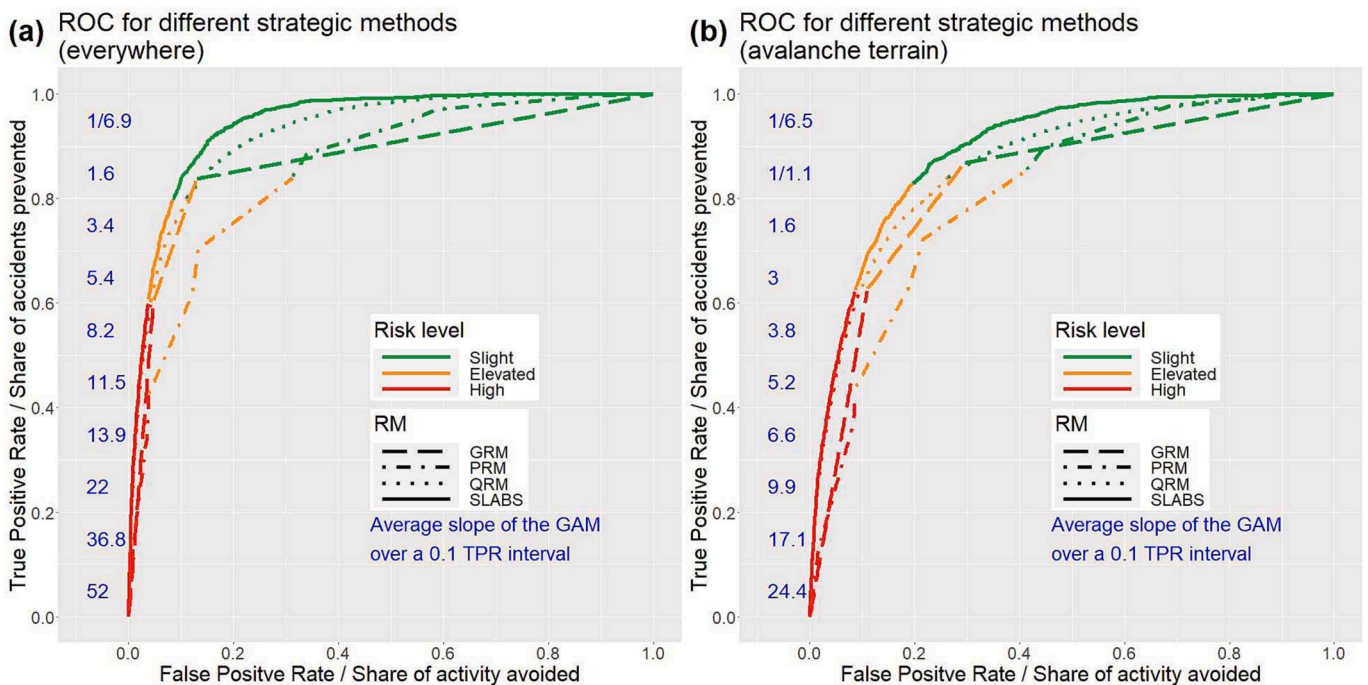


Fig. 4. ROC for the GRM, PRM, QRM and SLABS. The blue numbers on the left are the average slopes of the ROC of the SLABS method over 0.1 TPR intervals. They represent the relative risk between points in the interval and the average risk over all data (left) or in avalanche terrain (right). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

QRM were set such that 60% and 80% of accidents are prevented when points with *high* respectively *elevated* risk are avoided. For similar prevention rates as the GRM, the QRM offers more freedom. For the SLABS method the same thresholds were adopted. It offers more freedom of movement than the other methods at every prevention rate.

Choi (1998) gives an interesting interpretation of the slopes of the ROC. There is the slope in a particular point corresponding to a specific probability p that a data point is an accident. This slope corresponds to the likelihood ratio (LR in Eq. (5)) between the probability of a value p among the accidents divided by the probability of a value p among the travel points (Eq. (5)). Hence, points in the lower left corner of the ROC are much more likely to be accident points, while points in the upper right corner are much more likely to be travel points.

$$LR(p) = \frac{Pr(p|A=1)}{Pr(p|A=0)} = \frac{\text{Probability on } p \text{ among accidents}}{\text{Probability on } p \text{ among travel points}} \quad (5)$$

What is true for a single point on the ROC is also true for an interval. Assume that the probability p that a point is an accident lies in an interval $[p_l, p_h]$. Then the slope of the ROC over this interval is the likelihood ratio between $p \in [p_l, p_h]$ among accidents and $p \in [p_l, p_h]$ among travel points (Eq. (6)). This interpretation is more appropriate for the piece wise straight ROCs of the GRM and PRM.

$$LR(p_l \leq p \leq p_h) = \frac{Pr(p_l \leq p \leq p_h|A=1)}{Pr(p_l \leq p \leq p_h|A=0)} = \frac{\text{Probability on } p \in [p_l, p_h] \text{ among accidents}}{\text{Probability on } p \in [p_l, p_h] \text{ among travel points}} \quad (6)$$

In Fig. 4 the slope of the SLABS model in each TPR interval of 0.1 is shown in the column on the left. The data points with the highest risk scores are 57.1 times likelier to be accidents than travel points. The points with the lowest risk scores are 7.5 times less likely to be accidents than travel points. Considering these numbers gives more detailed information on the risk compared to the risk categories (*slight*, *elevated*, *high*).

5. Discussion

5.1. The model covariates

The *slope angle* is the most important covariate and was investigated in great detail. Different covariates describing the slope angle were tested: the local slope angle (SA), the maximum slope angle in different smaller or larger areas around a point (MSAx) and mixes of SA and MSAx depending on the danger level (SA_{-jx} and SA_{-p}). Also in practice a wider slope area is considered at higher danger levels because remote triggering becomes more likely. We found that The MSA40 is a better predictor than MSA70, MSA100 or MSA150. This finding allows to define more precisely the rather vague concept of *relevant slope*. Considering the slope angle in a radius of about 40 m seems appropriate. Though we have to be cautious at level 4-High. There are not enough data to determine the size of the area that has to be considered. Our analysis also confirms the strong increase in risk in the interval from 30° to 40°, corresponding to almost twice the risk ratio between two consecutive danger levels. The GRM and PRM use a similar increase, but for PRM with a risk ratio of 2 instead 4.3. Also the increase just below 30° is important. This is ignored by the GRM and PRM. An analysis of the Pearson residuals against MSA40 shows no trends (see Appendix A.7), indicating that the smoother captures the evolution of $\logit(risk)$ well. Also the residuals per danger level are free of trends, indicating that the smoother is valid for all danger levels. An analysis of the Pearson residuals of SA, SA_{-jx} and SA_{-p} shows some small trends but the residuals remain mostly within the $[-1, 1]$ interval. Also the Pearson residuals of the differences $dMSAx = MSAx - SA$ do not show relevant trends.

The *danger level* shows an almost linear relation with $\logit(risk)$. The risk ratio between subsequent levels is 4.3. This is about the double of

what the PRM assumes. This is not problematic for the performance of the PRM as long as the effects of other factors are properly scaled. For the slope angle we saw that the risk ratios between 30° to 40° are well represented but not below 30°. The validity of the model at level 4-High is weak. Only 9 accidents (0.74%) and 0.12% of the travel data occur at level 4-High. That is why the uncertainty at level 4-High is relatively big (see Fig. 2c). Moreover, spontaneous avalanches also increasingly become a danger. But these are not taken into account in our approach. At level 4, we therefore have a weak model. Before implementation in operational tools as the *skitourenguru* website, routes at level 4 have to be checked carefully.

The effect of the *core zone* is taken into account with two covariates: Delta critical Elevation (DCE) and the aspect overlapping fraction (AOF). The relation between DCE and $\logit(risk)$ could be approximated well with a smoother that rises almost linearly until 500 m above the critical elevation and then flattens off. With a risk ratio of 5.3 [3.9, 7] between -500 and 0 m and 2.7 [2.3, 3.1] between 0 and 500 m the impact of elevation is bigger than assumed in the GRM or PRM. The GRM assumes the equivalent of only one danger level between inside and outside the core zone while the risk ratio between 500 m above and 500 m below the critical elevation is 14.2. The effect of the critical aspects on the other hand is about 80% the effect of a danger level. This is in strong contrast with the PRM that gives a lot of weight to the relative risk between aspects: a factor 2 (or the equivalent of a danger level) less outside the aspects NW-N-NE and a factor 3 less outside the aspects WNW-N-ESE. This was already shown by Winkler et al. (2021) with univariate analysis on the same data set. Hence, our analysis refines the level-one rule giving more weight to the effect of elevation and less to the effect of aspects.

5.2. Covariates not in the model

This section discusses the covariates that were not included in the model. For these covariates trends in the Pearson residuals were checked.

The *avalanche problem* could not be included in the model because it is only available since the season 2012–2013. No trends were found in the Pearson residuals plotted against the avalanche problem. This means that the danger level holds the key information about the stability of the snow pack. This contrasts with the fact that Landrø et al. (2020a) observed that professionals pay more attention to the avalanche problem than to the slope angle and the danger level. We found that the latter two are the most informative ones. Therefore we would like to encourage professionals to also consider the statistically important parameters of danger level and slope angle in a first step, and only then deviate from this starting value based on other observations. The most frequent avalanche problems are *fresh snow*, *wind-drifted snow*, and *old snow*. This means that the model is mainly valid for these conditions. Very few accidents happen due to wet avalanches, hence the model will not be particularly good in these conditions. The same is true for glide snow avalanches but they are almost never triggered by skiers.

The Pearson residuals of the *month of the year* show no trends. This means that the avalanche warning service predicts the stability consistently throughout the winter season.

It is known that *frequently travelled slopes* are safer. If there are already many tracks on a slope, the probability of triggering an avalanche decreases per run. In addition, slopes that are regularly skied (fashionable tours) often have a more favourable snow pack structure. So, from a theoretical point of view, many ski runs should reduce the risk. We tried to capture this effect with a travel density calculated as the number of points in the neighbourhood of an accident or travel point. Unfortunately, there are not enough travel data to estimate how frequently travelled locations are. Since we have no information about the presence of tracks, our model gives risk ratios for an unknown average travel activity on the routes in the data set.

The PRM considers South and West facing slopes safer than North

and East facing ones. This effect is not found in the data. The covariates *NORTH* and *EAST* were not selected. The PRM was derived analysing accident counts per aspect without taking into account travel activity. Usually North facing slopes have better snow and are skied more frequently. As a consequence there are more accidents.

The covariate *DIST_RIDGE*, the *distance to the nearest ridge*, would have been selected but was not included in the model. It is too correlated to *DCE*: the higher you climb the closer you get to ridges (see Table A.4). When this covariate was included it obfuscated the effect of *DCE*. This illustrates a typical disadvantage of an observational study like the present one. The only way to separate the effects of *DIST_RIDGE* and *DCE* is doing a controlled experiment where one covariate is kept constant while the other one is varied and vice versa. This is difficult in the context of avalanches and would require many experiments due to the low base rate.

The *terrain fold and curvature (FOLD and PLANC)* were tested but not selected. To reduce the risk it is recommended to use ridges or convex terrain rather than gullies or concave terrain. But in the data no significant difference in risk could be found.

The *forest density (FD)* was not selected. This does not mean that forestation has no effect. The forest density is strongly correlated with the elevation and *DCE*. In the Alps the tree line lies about 2200 m. This altitude is close to the common critical elevations of the avalanche forecast. The strong effect of *DCE* is partially due to forestation but both effects cannot be separated from each other due to correlation. Forestation is further indirectly included in the model by the covariate *MSA40*.

The *maximum fall down velocity (FD_MAXV)* which is a measure for the potential consequences of an avalanche was included in the model selection. This covariate with smoother would have been selected right after *MSA40* but was not included in the final model. This decision was taken after considering the arguments in favour and against its inclusion. In favour of inclusion is of course the low BIC of models with *s(FD_MAXV)*. Also the AUC of a model including *FD_MAXV* with a smoother is 0.009 higher than the final model. As a consequence also the Pearson residuals of the final model show strong trends with *FD_MAXV*. Fig. 5 shows the Pearson residuals of *FD_MAXV*. They are negative when *FD_MAXV* is lower than 30 m/s, which means that the model overestimates the risk for accidents with potentially less severe

consequences. Above 30 m/s or accidents with potentially severe consequences the model underestimates the risk. This could be due to a reporting bias. Almost all accidents with severe or fatal consequences are reported to the Swiss avalanche warning service (SLF) while for less severe accidents there is an under-reporting. A reason against inclusion is the strong correlation with all slope angle variables (see Table A.4). When included, *s(FD_MAXV)* takes over a part of the slope angle effect and increases its standard errors on the smoother of *MSA40*. This was confirmed by comparing the final model with a model that additionally includes *s(FD_MAXV)* (see Fig. A.15). In the latter model the slope of the smoother of *MSA40* decreases and the standard error of the smoother increases. Finally *FD_MAXV*, says something about what happens after an avalanche has been triggered. Our objective is to build a model that connects the different causes of an avalanche. Including *FD_MAXV* would sin against causality. Considering these arguments we chose to leave *FD_MAXV* out of the model.

In the avalanche data set no information is available on the type of avalanche. An avalanche can be a dry or wet and a slab or point release (loose snow). However, nearly exclusively dry slab avalanches are triggered by backcountry skiers (Schweizer and Lüscher, 2001). If information were available, the model should be trained on dry slab avalanches only. This is a weakness of the study but we assume it has little impact on the results because of the predominance of dry slab avalanches.

5.3. On the thresholds used in the SLABS method

Thresholds were set to achieve prevention rates of 60% and 80% when avoiding high and elevated risk conditions, respectively. These values also correspond to the typical avoidance rates of other probabilistic methods analysed by McCammon and Haegeli (2007). If probabilistic methods are not accepted it is rather because of their perceived excessive avoidance than because of the remaining accidents in the green area. We have therefore decided to use the improved discriminatory power of SLABS not for fewer accidents with the same level of avoidance, but for more freedom of movement with the same preventive effect. Other areas of life, where absolute safety can only be achieved through total abstinence, could serve as an inspiration to set thresholds on the acceptable risk. However, the accepted risk is not a constant but a question of social acceptance. Efforts are still being made in Switzerland to reduce the risk of road traffic accidents even though the number of fatalities reduced tenfold since 1971 (Federal Statistical Office (BFS), 2023). We know that around 2010, the risk of dying in an avalanche on an average day of ski touring was around $8.7 \cdot 10^{-6}$ (Winkler et al., 2016). This would allow us to specify the risk of death, but not the risk of injury due to incomplete reporting. Moreover, because humans cannot deal with small probabilities (Kahneman et al., 2021), such numbers would not contribute to a meaningful handling of risks. It is neither possible to calculate a value for the acceptable risk because we cannot express the value of a fulfilling leisure activity or a person's life in monetary terms. We have therefore opted to stick to what is customary today for probabilistic methods.

6. Conclusions

In this paper we presented a new probabilistic method to assess avalanche risk ratios on backcountry ski tours: SLABS (Screening the Likelihood of Avalanche risk for Backcountry Skiing). A Generalised Additive Model (GAM) was fitted on a unique data set of both backcountry travel data and accidents. The model considers slope angle, danger level, elevation and aspect. Limits for *high* and *elevated* risk are set in such a way that accident prevention rates of respectively 60% and 80% are achieved when *high* or *elevated* risk terrain is avoided. These limits correspond to typical prevention rates of probabilistic methods, measured in McCammon and Haegeli (2007).

We were able to quantify the effect of four covariates on avalanche

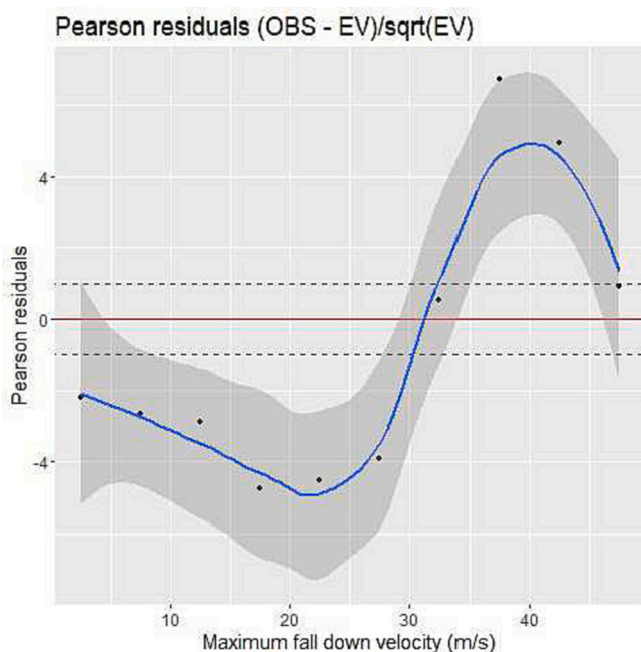


Fig. 5. Pearson residuals for the maximum fall-down velocity.

risk ratios: the non-linear relation with the slope angle, the risk ratios between danger levels, and the effect of elevation and aspect with respect to the core zone. We demonstrated that the risk varies about 5 orders of magnitude between the safest and most riskiest conditions. Hitherto the importance of individual risk factors had never been quantified.

These factors were already used by the GRM and PRM but not combined in the most optimal way. Especially the effect of aspect is overrated by the PRM while the effect of elevation with respect to the critical elevation is underrated. The size of *relevant slope* in which the maximum slope angle has to be considered could be quantified too. A control radius of about 40 m resulted to be most appropriate.

This paper presents the first complete analysis of the performance of existing reduction methods. In [McCammon and Haegeli \(2007\)](#) prevention rates for several methods were calculated but the False Positive Rate was ignored because travel data were not available. Because this study combines travel and accident data, for each probabilistic method the share of terrain that has to be avoided (False Positive Rate) to guarantee a desired accident prevention rate could be determined. The more freedom a method gives for a given prevention rate the better it is. The GRM is simpler and offers more freedom of movement for similar prevention rates than the PRM. Though the limit between *elevated* and *high risk* used by the PRM allows more freedom of movement at a lower risk prevention rate than the GRM. The QRM scores better than the GRM and the PRM. The SLABS method based on a GAM gives the most freedom of movement for any given accident prevention rate.

SLABS is not only the most accurate probabilistic method, but also the one with the best scientific basis. In order to make SLABS applicable in practice, a front-end is needed, such as already exists with www.skitouren guru.ch. As probabilistic method, SLABS remains an initial estimate for a single slope. During a ski tour, it should be supplemented with an independent assessment of the avalanche danger on site. Observations in the terrain can tip the decision in either direction: continue or return. These local observation should be important factors that are not included in the model, such as:

- Recent skiing activity on the slope. Since we had no information about the presence of tracks, our method gives risk ratios for an average travel activity. If many tracks are present on a slope it is probably safer than the model indicates. If no tracks are present the slope is probably less safe than the model indicates.
- Danger signs that do not match with the current danger level. E.g. recent spontaneous avalanches, whumpf sounds and cracks at danger level 1-Low or 2-Moderate where these are unusual.
- Local variation of the snow cover. The avalanche forecast is not able to predict local variations in the snow cover and stability. E.g. local accumulations of wind-drifted snow increase the risk while on a blown-off slope the risk is lower.
- Safe terrain features like ridges, dense forests, human made artefacts or, the other way round, terrain traps.

Unfortunately we cannot quantify these effects. Only a qualitative assessment can be made about the avalanche risk from such local observations. Some observations might indicate an increase and others a decrease in risk. Combining the risk assessment of SLABS with local observations is still a big challenge. Assessing the avalanche risk is still associated with big uncertainties. However, with this study we believe to have quantified better some of the most important factors. By implementing the SLABS method in the website www.skitouren guru.ch this knowledge is made available to the public which hopefully contributes to everyone's safety.

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Author statement

We the undersigned declare that this manuscript is original, has not been published before and is not currently being considered for publication elsewhere.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We understand that the Corresponding Author is the sole contact for the Editorial process. He/she is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs.

CRediT authorship contribution statement

Bart Degraeuwe: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Günter Schmudlach:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Investigation, Data curation, Conceptualization. **Kurt Winkler:** Writing – review & editing, Writing – original draft, Supervision, Methodology. **Jochen Köhler:** Writing – review & editing, Writing – original draft, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data on accidental avalanches and backcountry touring activity and accidents were extracted from the ARPD ([Schmudlach, 2021a](#)). According to the privacy agreement with the owners of the GPS tracks, it is not possible to publish the travel points. However, interested researchers can request a table containing all parameters for all travel points, except the geographical coordinates, for the exclusive usage of research verification. In order to enable the verification of the data table, eventual subscribers can request the geographical coordinates for 100 random lines out of the data table.

An R script that reproduces [Fig. 3](#) and allows to calculate the *logit risk* and risk ratios for different combinations of covariates is provided as supplementary material.

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Appendix

A.1. Probabilistic methods

The *Graphical Reduction Method* (GRM) combines slope angle and danger level to determine a discrete risk score: *slight*, *elevated* or *high*. At danger levels 1-Low and 2-Moderate the slope angle in the vicinity of the track, within a radius of 20 m must be used. At danger level 3-Considerable the maximum slope angle of the entire *relevant slope* must be used. At danger level 4-High also run-out zones must be considered. The GRM does not specify clearly how to determine the *relevant slope*. In [Section 2](#) and the [Appendix A.3](#) this issue is discussed in more detail. On slopes with aspect or elevation not mentioned in the forecast one can assume the next lower danger level. This is also called the one-level rule ([WSL Institute for Snow and Avalanche Research SLF, 2021](#)). The risk is considered *slight* on slopes below 40° at level 1-low, on slopes below 35° at level 2-Moderate and on slopes below 30° at level 3-Considerable. On slopes up to 5° steeper at the same danger level, the risk is considered *elevated*. Travelling in slopes more than 5° steeper at the same danger level is not recommended because the risk is *high*. A detailed description and graphic representation can be found in the leaflet published by [WSL Institute for Snow and Avalanche Research SLF et al. \(2018\)](#).

The *Professional Reduction Method* (PRM) was the first heuristic method, proposed in the 1990s by [Munter \(2017\)](#). A so-called residual risk is calculated by dividing a danger potential (*DP*) by a product of reduction factors (*RF*) as in Eq. (A.1a). The danger potential is equal to 2 elevated to the danger level (*DL*) given by the avalanche forecast. A residual risk below or equal to 1 is considered acceptable. There are three classes of RFs. The first class RF depends on the slope angle (*SA*) (see Eq. (A.1b)). The use of the slope angle in the vicinity of the track or in the entire *relevant slope* is the same as for the GRM. At level 3-Considerable a first class RF is compulsory. The second class RF depends on the aspect of the slope (see Eq. (A.1c)). The Northern quadrant and Northern hemisphere are more dangerous. Outside the core zone of the avalanche forecast, i.e. outside the mentioned aspects or elevations, the risk is lower. Also frequently skied slopes are considered safer. Second class RFs cannot be used when the avalanche problem is wet snow. The third class RF depends on the group size and the use of safety distances (see Eq. (A.1d)). In the ROC curve we consider the residual risk of 1 as the border between *elevated* and *high* risk. We introduced an additional threshold at a residual risk of 1/3 and define it as the border between *slight* and *elevated* risk in the ROC curves. This threshold corresponds to the RF_3 for a small group using safety distances. This precautionary measures is also recommended for the orange area of the GRM and the QRM. We therefore use the “small group with safety distances” factor to define an orange area.

$$\text{residual risk} = \frac{2^{DL}}{RF_1 \cdot RF_2 \cdot RF_3} \quad (\text{A.1a})$$

$$\begin{aligned} RF_1 &= 2 \text{ if } 35^\circ < SA < 40^\circ \text{ or} \\ &= 3 \text{ if } SA \approx 35^\circ \text{ or} \\ &= 4 \text{ if } SA < 35^\circ. \end{aligned} \quad (\text{A.1b})$$

$$\begin{aligned} RF_2 &= 2 \text{ if Aspect outside NW – N – NE or} \\ &= 3 \text{ if Aspect outside WNW – N – ESE or} \\ &= 4 \text{ if outside core zone of the avalanche forecast.} \\ &= 2 \text{ if frequently skied.} \end{aligned} \quad (\text{A.1c})$$

$$\begin{aligned} RF_3 &= 2 \text{ if small group (} \leq 4 \text{) or big group keeping distances or} \\ &= 3 \text{ if small group keeping distances.} \end{aligned} \quad (\text{A.1d})$$

The *Quantitative Reduction Method* (QRM): This is the method proposed by [Schmudlach et al. \(2018b\)](#) and currently implemented in [www.skitour enguru.ch](#). It combines two indicators into a relative risk: the danger indicator (*DI*) and the terrain indicator (*TI*). The *DI* describes the danger predicted by the avalanche forecast. It is a spatially interpolated version of the danger level taking into account the one-level rule. The *TI* indicates how appropriate the terrain is to trigger an avalanche. The *TI* depends on the slope angles on the RSA and on the size of the RSA. The QRM was derived from accident and travel data described in [Section 2](#). Accident and travel data were binned according to their *DI* and *TI*. Then a smoothing kernel was applied ([Hastie et al., 2009](#)). The ratio of smoothed accident and travel point density per *DI-TI* bin determines the relative risk. The risk boundaries between *slight-elevated* and *elevated-high* were set to achieve accident prevention rates of 80% and 60%, respectively.

The risk scores provided by these methods allow comparing different conditions with a reference situation for which risk is considered acceptable.

A.2. Covariate properties

This annex gives an overview of the properties of the covariates considered for the model development. [Table A.3](#) gives the mean, median, minimum, maximum and standard deviation of each continuous covariate both for accidents and non-accidents. [Fig. A.6](#) shows the probability density function of each continuous covariate both for accidents and non-accidents. [Fig. A.7](#) shows the distribution of the predicted danger level in the avalanche bulletin for both accidents and non-accidents. [Fig. A.8](#) shows the share of data points in which a given avalanche problem is mentioned both for accidents and non-accidents. The sum of the shares per accident or non-accident are bigger than 1 because more than one problem can be mentioned. [Fig. A.9](#) shows the distribution of the accident and non-accident data over the seasons. The availability of travel data increased strongly from season 05–06 to 12–13 due to the proliferation of smartphones.

Table A.3

Mean, median, minimum, maximum and standard deviation of accident and non-accident data for all continuous covariates.

Covariate	Accident	Mean	Median	Minimum	Maximum	SD
AOF (–)	No	0.74	1	0	1	0.39
AOF (–)	Yes	0.9	1	0	1	0.26
ASPECT (°)	No	176	165	0	360	110
ASPECT (°)	Yes	179	167	0	360	119

(continued on next page)

Table A.3 (continued)

Covariate	Accident	Mean	Median	Minimum	Maximum	SD
CE (m.a.s.l.)	No	1614	2000	−3000	3000	914
CE (m.a.s.l.)	Yes	1969	2000	−3000	3000	504
DCA (°)	No	103	180	−135	180	90
DCA (°)	Yes	118	180	−135	180	76
DCE (m)	No	−21	−56	−2379	3543	595
DCE (m)	Yes	393	393	−880	3348	402
DIST_RIDGE (m)	No	704	579	0	3000	601
DIST_RIDGE (m)	Yes	250	134	0	3000	295
ELE (m.a.s.l.)	No	2041	2018	200	4624	592
ELE (m.a.s.l.)	Yes	2439	2457	1120	3890	443
FD (%)	No	11	0	0	100	25
FD (%)	Yes	3.2	0	0	86.5	13
FD_MAXV (m/s)	No	20	21	0	141	12
FD_MAXV (m/s)	Yes	34.1	35.2	0	67.4	8.6
FOLD Fold (°)	No	0	−2	−139	145	15
FOLD Fold (°)	Yes	−5.5	−6	−90.5	93.8	14.4
MSA100 (°)	No	28.8	28.9	0	86.6	10.8
MSA100 (°)	Yes	43.5	42.5	2.4	76.3	8.3
MSA150 (°)	No	29.8	29.9	0	86.5	10.4
MSA150 (°)	Yes	43.6	42.6	2.4	75.9	8
MSA40 (°)	No	26	25.8	0	86.8	11.2
MSA40 (°)	Yes	42.5	41.4	2.4	76.5	8.7
MSA70 (°)	No	27.9	27.8	0	86.8	11.1
MSA70 (°)	Yes	43.3	42.2	2.4	76.3	8.4
PLANC (−)	No	0.5	0.8	−100	100	22
PLANC (−)	Yes	5.4	4.3	−99.5	99.7	14.5
SA (°)	No	20	19.6	0	86	9.6
SA (°)	Yes	34.7	35.1	0	74	7.7
TI (−)	No	0.25	0.174	0	0.965	0.251
TI (−)	Yes	0.646	0.705	0	0.946	0.178

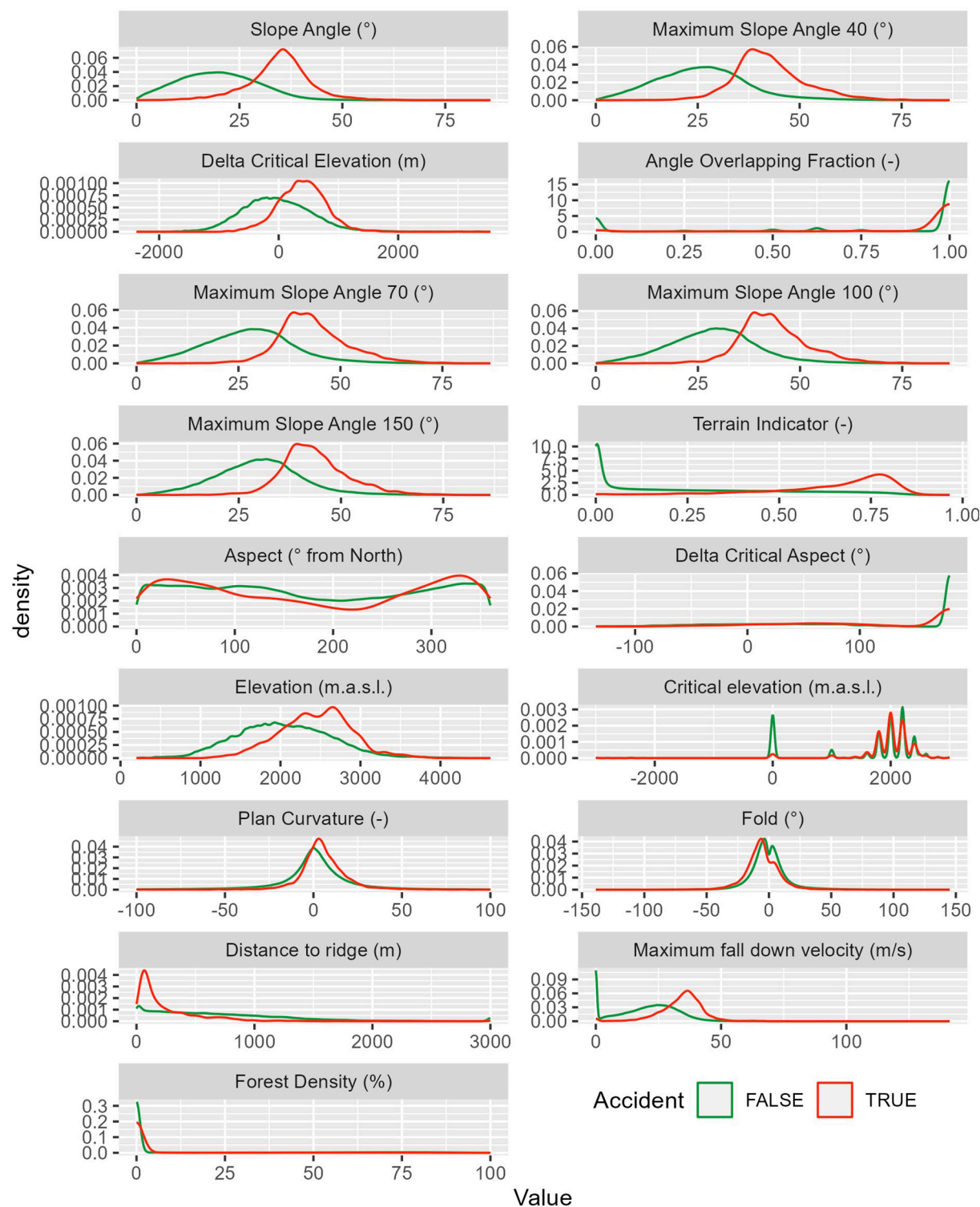


Fig. A.6. Probability Density Function of each continuous covariate for both accidents and non-accidents.

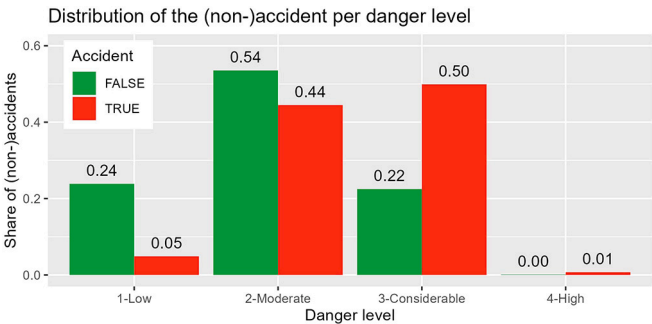


Fig. A.7. Danger level distribution for both accidents and non-accidents.

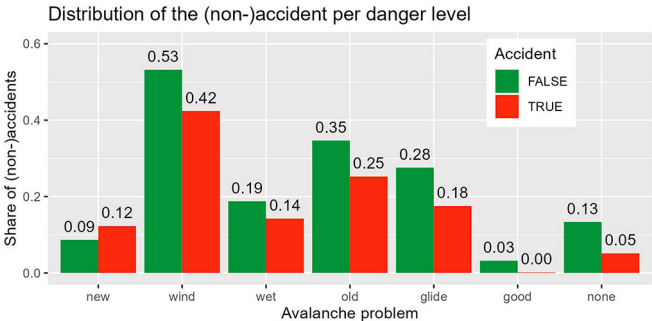


Fig. A.8. Share of the data in which a specific avalanche problem is mentioned for both accidents and non-accidents. This information is only available since season 2012–2013.

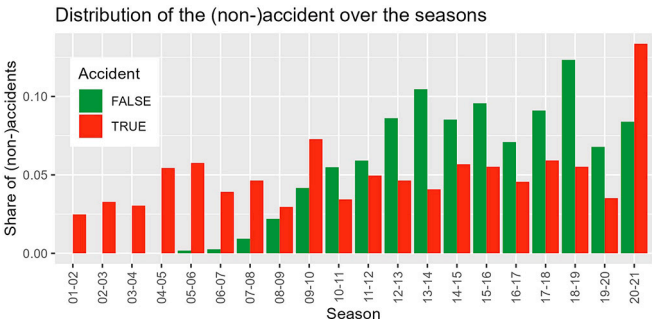


Fig. A.9. Share of the data per season for both accidents and non-accidents.

A.3. Definition of the Relevant Slope Area (RSA)

In the 1980s, when Munter (Munter, 2017) pointed to the *slope angle* as an important avalanche factor, he already made clear that geomorphological properties in a larger environment around the skier play a role. In order to apply the PRM users therefore have to identify the slope angle of the “steepest slope section”. The GRM specifies that, depending on the danger level, the slope angle has to be considered in the “vicinity of the track”, in the “entire slope” or even in the “entire slope including the run-out zone”. However the concept of the Relevant Slope Area (RSA) remained largely undefined.

In order to better understand the reflections of mountaineers Schmudlach et al. (2018a) conducted a survey among 26 avalanche experts. The participants had to delineate for 24 points in the terrain the slope area they considered relevant for their endangerment if standing at the points. Fig. A.10 shows the solutions of the experts (transparent-green) for two exemplary points (orange). The red polygon indicates the average RSA as proposed by the experts.

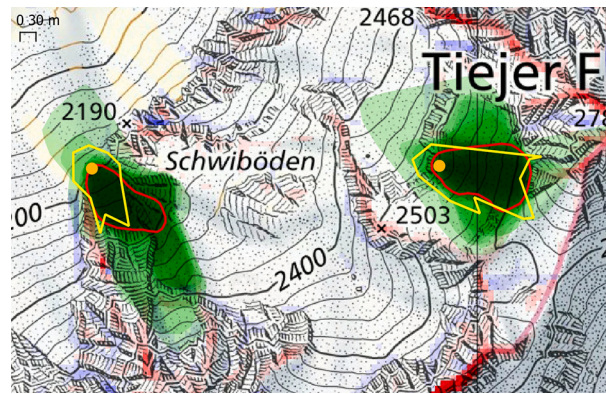


Fig. A.10. Relevant Slope Area (RSA) as defined by experts: Orange is the point to be assessed. The RSA suggested by experts are marked with transparent-green colour. The average RSA from the experts is outlined with a red colour. The RSA calculated by the algorithm is outlined with a yellow colour. Terrain folds are displayed with transparent blue (concave) and red (convex) colors. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

In the next step the algorithm of [Schmudlach and Köhler \(2016\)](#) was refined to emulate the RSA proposed by the experts. Its based on two steps:

1. A base form, consisting of two linked circle segments is aligned parallel to the gradient at the point (see [Fig. A.11](#)). The base form is controlled by the radius r_1 , r_2 and the angle α . These parameters are a function of the plan curvature (PLANC), the slope angle (SA) and a configuration parameter MSAR (Maximum Slope Area Radius). If $r_1 = r_2$ and $\alpha = 180^\circ$ the base form becomes a circle.
2. Avalanche experts typically refer to nearest terrain fold in any direction, when asked about the delimitation of the RSA. A terrain fold can be a ridge, the bottom line of a valley, a hillside toe or a slope edge. Terrain folds are defined by the maximum angle between 5 pairs of normal vectors placed oppositely on a circle with radius 10 m around the point. In the second step the base form is cropped, where a threshold of accumulated fold values is reached.

Summarised the RSA gets delimited by the next fold, the next forest or the base form. [Fig. A.11](#) gives an illustrative example. In [Fig. A.10](#) the RSA calculated by the algorithm (yellow) can be compared to the RSA proposed by the experts (red). Implementation code in C# is available in [Schmudlach \(2021b\)](#).

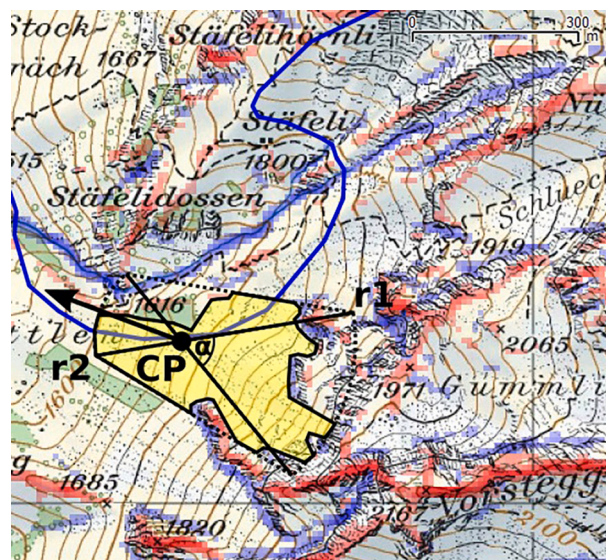


Fig. A.11. Relevant Slope Area (RSA) according to algorithm described in [Schmudlach and Köhler \(2016\)](#): The RSA is delimited by the next fold (red for convex folds respectively blue for concave folds), the next forest or the base form (defined by r_1 , r_2 and α). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

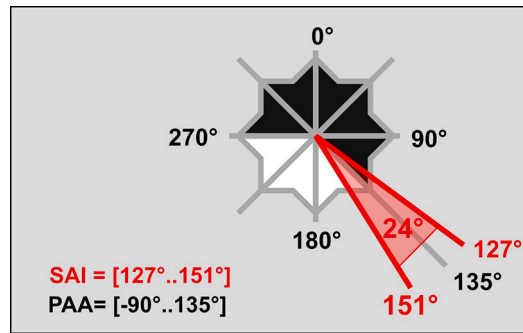


Fig. A.12. The Aspect Overlapping Fraction (AOF) is the fraction of the Slope Aspect Interval (SAI) that overlaps with the particularly affected (critical) aspects (PAA) mentioned in the avalanche forecast.

As soon as the RSA is known the MSAX can be derived by calculating the 85th percentile of the slope angles on the RSA. The control parameter x (40, 70, 100, 150) defines the applied MSAR value. MSAR is loosely linked to the size of the RSA. However, the MSAR is only a control parameter and should not be miss-interpreted as “remote triggering distance”.

Also the Aspect Overlapping Fraction (AOF) is calculated from the RSA. For a given location the aspects of all the points in its RSA are determined. The range of these aspects gives the Slope Aspect Interval (SAI). The avalanche forecast gives the Particularly Affected critical Aspects (PAA). The fraction of the SAI that overlaps with the in the PAA is the AOF. In the example of Fig. A.12 a third of the SAI overlaps with the PAA. Taking into account slope angles in a larger environment around the skiers position is a crucial concept when assessing avalanche terrain and already suggested by Munter (2017). However, the authors want to point out that the RSA proposed by Schmudlach and Köhler (2016) remains a subjective concept.

A.4. Correlation matrix of terrain parameters

In Table A.4 you find a correlation matrix of SA, MSA40, MSA70, TI, FD_MAXV, ELE, DCE, ATH, PLANC, FOLD, FD and DIST_RIDGE. All slope angle covariates are strongly correlated (> 0.70). DIST_RIDGE and ELE are strongly correlated (0.66). FD_MAXV and MSA40 are moderately correlated (0.39). FD and ELE are also moderately correlated.

Table A.4

Pearson correlation coefficients of the terrain properties.

	SA	MSA40	MSA70	TI	ELE	DCE	ATH	PLANC	FOLD	FD	FD_MAXV	DIST_RIDGE
SA	1.00	0.78	0.72	0.55	0.08	0.07	0.63	−0.01	−0.01	0.20	0.44	−0.17
MSA40	0.78	1.00	0.96	0.79	0.13	0.11	0.68	0.14	−0.28	0.03	0.39	−0.21
MSA70	0.72	0.96	1.00	0.84	0.15	0.12	0.68	0.16	−0.30	−0.02	0.38	−0.23
TI	0.55	0.79	0.84	1.00	0.20	0.16	0.76	0.15	−0.32	−0.13	0.36	−0.24
ELE	0.08	0.13	0.15	0.20	1.00	0.88	0.22	−0.04	0.06	−0.42	0.42	−0.66
DCE	0.07	0.11	0.12	0.16	0.88	1.00	0.20	−0.04	0.07	−0.38	0.39	−0.60
ATH	0.63	0.68	0.68	0.76	0.22	0.20	1.00	0.05	−0.13	−0.13	0.48	−0.27
PLANC	−0.01	0.14	0.16	0.15	−0.04	−0.04	0.05	1.00	−0.39	−0.00	−0.13	0.05
FOLD	−0.01	−0.28	−0.30	−0.32	0.06	0.07	−0.13	−0.39	1.00	0.00	0.24	−0.08
FD	0.20	0.03	−0.02	−0.13	−0.42	−0.38	−0.13	−0.00	0.00	1.00	−0.40	0.32
FD_MAXV	0.44	0.39	0.38	0.36	0.42	0.39	0.48	−0.13	0.24	−0.40	1.00	−0.46
DIST_RIDGE	−0.17	−0.21	−0.23	−0.24	−0.66	−0.60	−0.27	0.05	−0.08	0.32	−0.46	1.00

A.5. The effect of the smoother type

The type of smoother can have an impact on the shape of the smoother. The default smoother type of the R-mgcv package was used, a thin plate spline. For the smoothers of MSA40 and DCE the sensitivity of the result to the type of smoother was investigated. Fig. A.13 shows the smoother of MSA40 for 7 different types of splines. Where it matters, especially between 30 and 45°, the differences between the spline types fall within the uncertainty on the thin plate spline. The uncertainty near the edges of the range is more due to the lack of accidents.

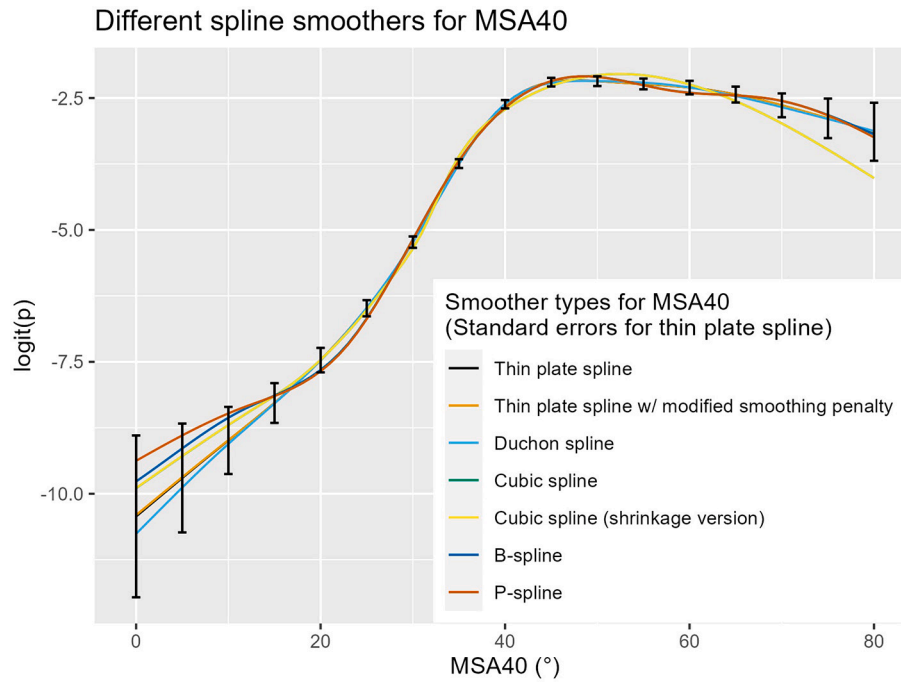


Fig. A.13. Smoother of *MSA40* for 7 different types of splines. The error bars correspond to the standard error of the thin plate spline.

Fig. A.14 shows the smoother of *DCDE* for 6 different types of splines (The Duchon spline had a very long calculation time). The differences between the spline types fall within the uncertainty on the thin plate spline. The higher uncertainty near the edges of the range is more due to the lack of accidents.

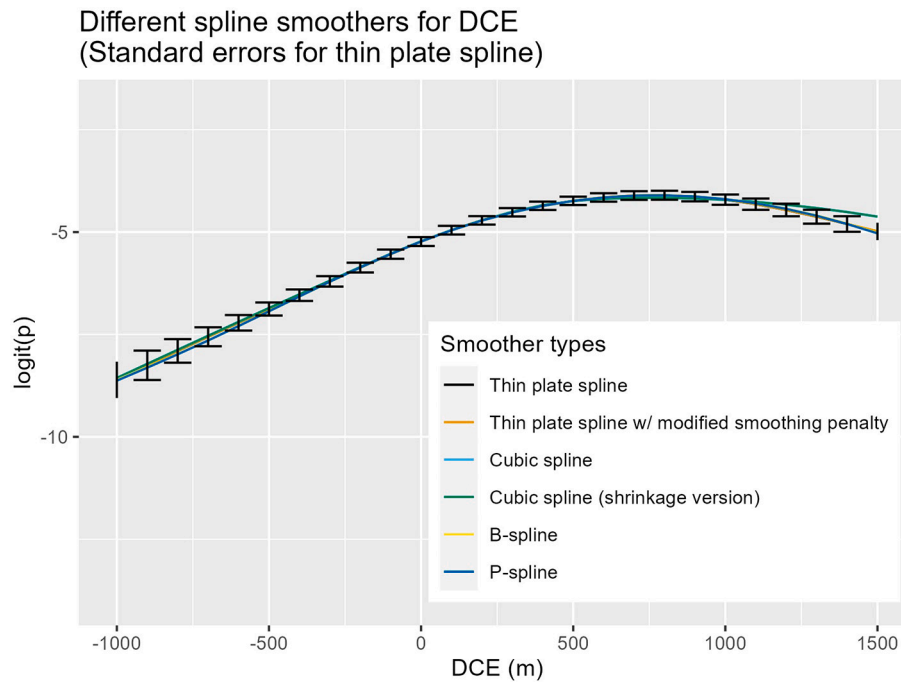


Fig. A.14. Smoother of *DCE* for 6 different types of splines. The error bars correspond to the standard error of the thin plate spline.

A.6. Smoother of *MSA40* of a model with or without $s(FD_MAXV)$

When a smoother of *FD_MAXV* is added to the final model (Eq. (4)) the standard errors of the smoother of *MSA40* increase and the strength of its effect decreases. This is a reason not to include *FD_MAXV* in the model despite the fact that the area under curve (AUC) improves by 0.009 (see Table A.5).

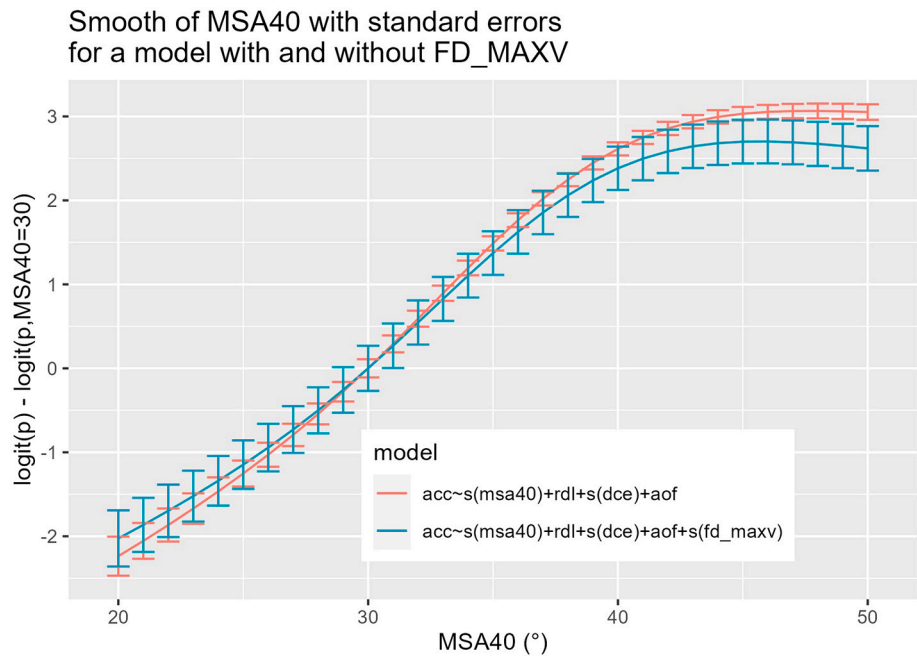


Fig. A.15. Smooth of MSA40 with standard errors for a model with and without $s(FD_MAXV)$. The smooths were set equal for $MSA40 = 30^\circ$ to make comparison easier.

Table A.5
Area under curve (AUC) of the 5 folds and their average for the final model and a model including $s(FD_MAXV)$.

Model \ AUC	Fold1	Fold2	Fold3	Fold4	Fold5	Mean AUC
$acc \sim s(msa40) + dl + s(dce) + aof$	0.933	0.943	0.949	0.946	0.945	0.943
$acc \sim s(msa40) + dl + s(dce) + aof + s(fd_maxv)$	0.943	0.950	0.958	0.956	0.953	0.952

A.7. Pearson residuals of the covariates in the model

Fig. A.16 shows the Pearson residual plots for the covariates in the model: MSA40, DL, DCE and AOF. Above each residual the number of accidents is reported. For MSA40 and DCE bins were chosen to have the same amount of accidents (50). Bins of equal with and slightly more or less bins give similar results. For DL there are only four bins. Because most values of AOF are 0 or 1, only four bins were taken. To verify that the Pearson residuals are normally distributed, normal quantile plots were added (see Fig. A.17).

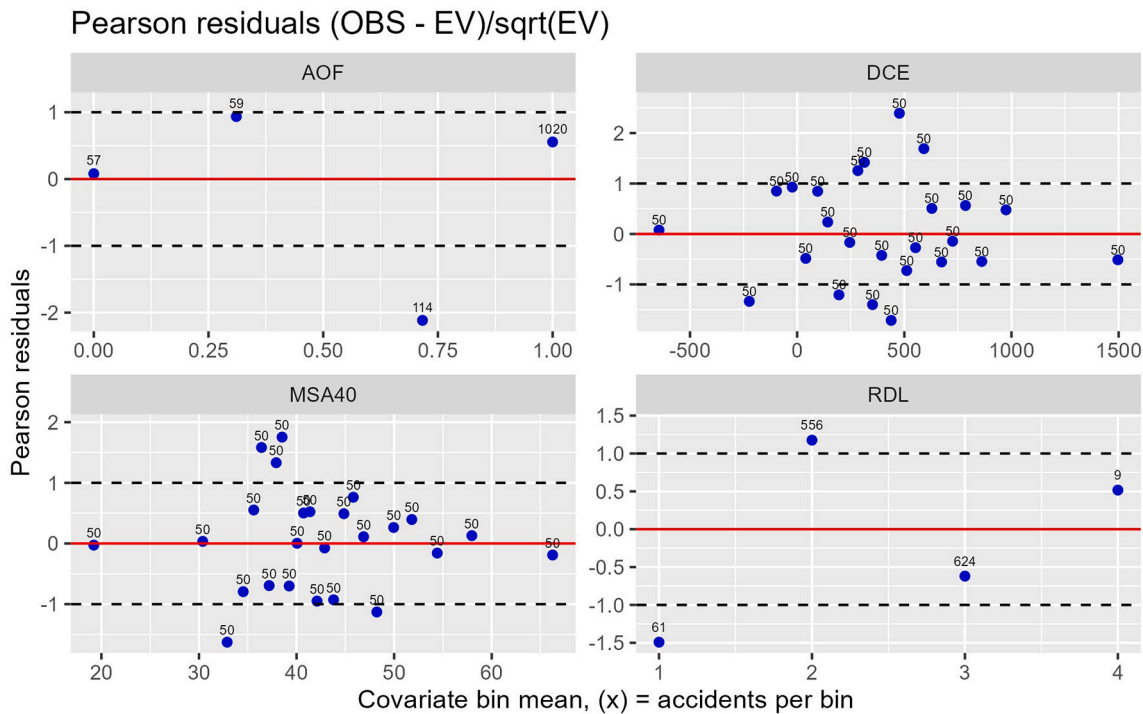


Fig. A.16. Pearson residuals for the four covariates in the model. The numbers above each residual indicate the number of accidents in the covariate interval for which the residual is calculated.

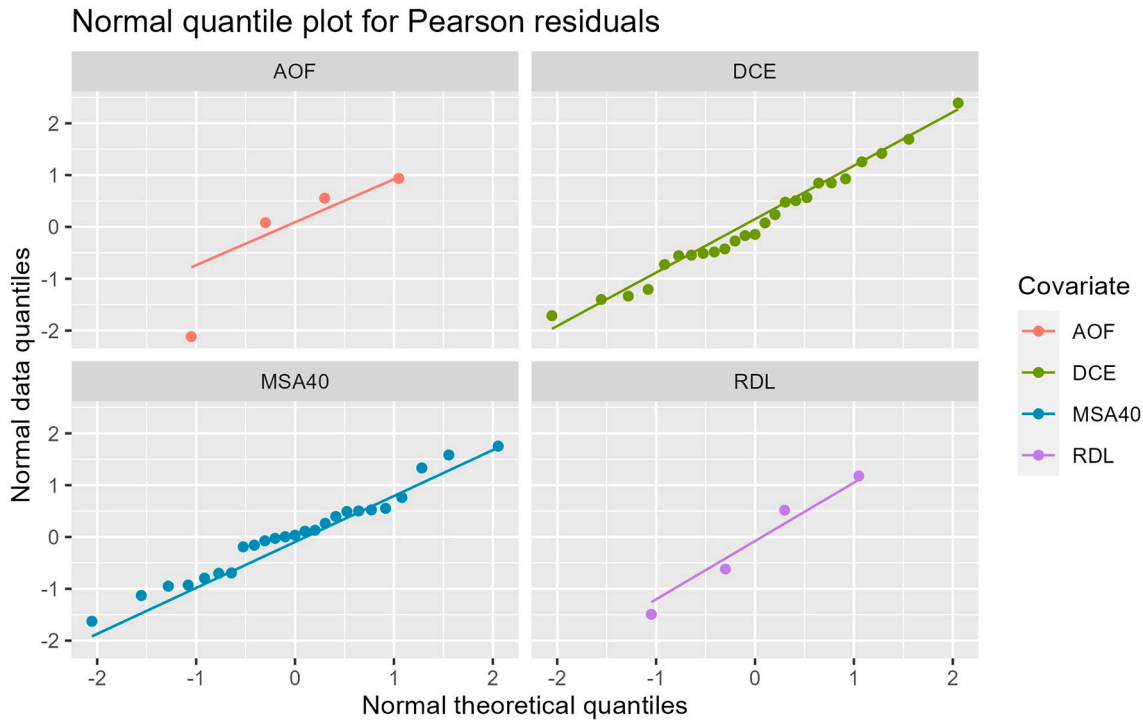


Fig. A.17. Normal quantile plot of the Pearson residuals of the four covariates in the model.

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